# YENEPOYA UNIVERSITY

## Project – High Level Design (HLD)

on

## Natural Language Interface for E-Commerce Infrastructure

(ShopEase E-Commerce Chatbot)

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# 1. Introduction

This document presents the High Level Design (HLD) of the Natural Language Interface for E-Commerce Infrastructure — a rule-based chatbot system built using Python 3. The system allows users to interact with an e-commerce support platform using everyday conversational text, without requiring human customer service agents.

The chatbot handles common customer queries including order tracking, payment options, return policies, shipping information, and discount offers. It is available in two versions — a Command Line Interface (CLI) built in Python, and a Web Interface built with HTML, CSS, and JavaScript.

## 1.1 Scope of the Document

This HLD document covers the following:

- The overall system architecture and how all components connect
- The process flow from user input to bot response
- The data model used to store keyword-response pairs
- The interface design for both the CLI and web versions
- Non-functional requirements — security and performance

This document does NOT cover the detailed internal logic of individual functions. That is documented in the Low Level Design (LLD) document.

## 1.2 Intended Audience

| Audience | Purpose |
|---|---|
| Professor / Evaluator | To assess the architecture and design choices made in the project |
| Developer (Student) | To use as a reference blueprint while building and extending the project |
| Industry Mentor | To review the technical approach and provide guidance |
| Future Developers | To understand the system before making enhancements |

## 1.3 System Overview

The ShopEase Chatbot is a Natural Language Interface built on a rule-based keyword matching engine. It accepts free-form text from the user, identifies the intent using string matching, and returns a pre-written response from its knowledge base.

| Property | Value |
|---|---|
| Project Title | Natural Language Interface for E-Commerce Infrastructure |
| Technology — Backend | Python 3 (chatbot.py, responses.py, utils.py) |
| Technology — Frontend | HTML5, CSS3, JavaScript (index.html) |
| Matching Technique | Basic String Matching using Python 'in' operator |
| Number of Topics Covered | 7 (Orders, Payments, Returns, Shipping, Offers, Support, Cancellation) |
| External Libraries Used | None — Python standard library only |
| Development Tool | Visual Studio Code (VS Code) |

# 2. System Design

The ShopEase Chatbot follows a simple 3-layer architecture — Presentation, Logic, and Data — with no external dependencies or network calls required.

## 2.1 Application Design

The system is divided into three logical layers:

| Layer | Description |
|---|---|
| Presentation Layer | Handles user input and displays output to the screen |
| Logic Layer | Keyword matching engine — finds the right reply for each input |
| Data Layer | Static in-memory knowledge base of keyword-response pairs |

## 2.2 Process Flow

The following is the step-by-step flow for every user message sent to the chatbot:

1. User launches the chatbot by running: python chatbot.py in the terminal
2. The welcome banner is displayed on screen by display_banner() in utils.py
3. The program enters a continuous while True loop — waiting for user input
4. User types a message and presses Enter
5. The raw input is passed to get_response() in responses.py
6. Input is normalized to lowercase and extra spaces are removed using .lower().strip()
7. A nested loop checks each keyword in RESPONSES using the 'in' operator
8. If a keyword is found inside the input — the matching response is returned immediately
9. If no keyword matches after checking all entries — the FALLBACK message is returned
10. The response is formatted and displayed via format_bot_reply() in utils.py
11. The loop returns to step 3 and waits for the next message
12. When user types 'exit' or 'quit' — the loop breaks and the program closes

## 2.3 Information Flow

Information travels in a single direction — from user input through the logic layer to the screen output. The diagram below shows how data moves between components:

Terminal Version (Python):

| From | To | Data Transferred |
|---|---|---|
| User Keyboard | chatbot.py | Raw text string entered by the user |
| chatbot.py | responses.py | Normalized (lowercase, stripped) input string |
| responses.py | chatbot.py | Matched response string or FALLBACK |
| chatbot.py | utils.py | Response string to be formatted |
| utils.py | Terminal Screen | Final formatted text printed to the screen |

Web Version (JavaScript):

| From | To | Data Transferred |
|---|---|---|
| HTML Input Box | sendMessage() | Raw text typed by the user |
| sendMessage() | getResponse() | Normalized input string |
| getResponse() | addMessage() | Matched response string |
| addMessage() | Browser Screen | Chat bubble rendered with reply text |

## 2.4 Components Design

The project is made up of 4 files, each with a clearly defined single responsibility:

| Component / File | Responsibility | |
|---|---|---|
| chatbot.py | Main entry point. Contains run_chatbot() — the while True input loop and command routing. | |
| responses.py | Knowledge base. Contains the RESPONSES list and get_response() matching function. | |
| utils.py | Display module. Contains display_banner(), display_help(), and format_bot_reply(). | |
| index.html | Web version. A self-contained file with HTML structure, CSS styling, and JavaScript logic. | |

Internal function summary:

| Function | File | Purpose |
| --- | --- | --- |
| run_chatbot() | chatbot.py | Starts and manages the continuous chat loop |
| get_response(input) | responses.py | Returns a matching response string for user input |
| display_banner() | utils.py | Prints the welcome header to the terminal |
| display_help() | utils.py | Prints the formatted command reference table |
| format_bot_reply(msg) | utils.py | Formats and prints the bot reply with indentation |
| sendMessage() | index.html | Handles the send button click on the web interface |
| getResponse(input) | index.html | JavaScript version of the keyword matching engine |
| addMessage(text,who) | index.html | Creates and adds a chat bubble to the screen |

## 2.5 Key Design Considerations

| Design Decision | Reason |
|---|---|
| Single HTML file for web version | Easy to share and open — no web server or installation needed |
| Static in-memory data — no database | Keeps the project lightweight and simple for a beginner project |
| 3 separate Python files (modular design) | Each file has one job — easier to understand, test, and extend |
| while True loop in chatbot.py | Keeps the conversation running until the user explicitly exits |
| .lower().strip() normalization | Makes matching case-insensitive and tolerant of accidental spaces |
| First-match-wins strategy | Simple and fast — no ambiguity, predictable behaviour every time |
| textContent instead of innerHTML (JS) | Prevents XSS (Cross-Site Scripting) attacks in the web version |

## 2.6 API Catalogue

This project does not consume any external APIs. All functions serve as internal APIs between the three Python modules. The table below lists each internal API:

| Function (Internal API) | Input | Output |
|---|---|---|
| get_response(user_input) | String — raw user text | String — matched response or FALLBACK |
| display_banner() | None | None — prints to terminal |
| display_help() | None | None — prints command table |
| format_bot_reply(message) | String — response text | None — prints formatted reply |
| getResponse(userInput) [JS] | String — user text | String — matched response |
| addMessage(text, sender) [JS] | String, String | None — renders chat bubble |

# 3. Data Design

The system uses a single in-memory data structure. No external database, file storage, or cloud service is used.

## 3.1 Data Model

All chatbot knowledge is stored in a Python list of dictionaries named RESPONSES, defined in responses.py. Each dictionary represents one topic category:

| Field | Type |
|---|---|
| keywords | List of strings |
| response | String (multi-line) |
| FALLBACK | String constant |

Example entry from the RESPONSES list:

```
{ "keywords": ["order status", "track order", "show status"],

  "response": "ORDER STATUS\nYour order #ORD-20481 is: OUT FOR DELIVERY..." }
```

## 3.2 Data Access Mechanism

Data is accessed using a sequential linear search — no indexing, querying, or sorting is used:

1. Iterate through RESPONSES list from index 0 to end
2. For each entry, check all keywords using Python's 'in' operator
3. Return the first matching entry's response immediately
4. If no match found after the full loop — return FALLBACK string

| Property | Value | |
|---|---|---|
| Algorithm | Sequential Linear Search | |
| Time Complexity | O(R x K) — R = response entries (7), K = keywords per entry (~5) | |
| Space Complexity | O(1) — no extra memory used during the search | |
| Worst Case | 35 checks (7 entries x 5 keywords) — completes in under 1ms | |

## 3.3 Data Retention Policies

The current version does not retain any data after the session ends. All conversation data exists only in RAM during the active session and is permanently lost when the program closes. No log files, databases, or cookies are written.

Future improvement: Chat history could be saved to a .txt file using Python's built-in open() and write() functions, allowing conversations to be reviewed after the session.

## 3.4 Data Migration

Not applicable for the current version. The system uses hardcoded static data with no external storage. Future versions that connect to a real database (e.g., SQLite or MySQL) would require a migration plan to move the RESPONSES list into structured database tables.

# 4. Interfaces

The chatbot is available through two distinct interfaces — a terminal-based CLI and a browser-based web interface. Both use the same keyword matching logic.

### 4.1 Terminal Interface (CLI)

| Interface Element | Description |
|---|---|
| Input Method | Python input() — waits for user keyboard text and Enter key |
| Output Method | Python print() — displays text in the terminal window |
| Welcome Screen | ASCII banner printed once on startup by display_banner() |
| Help Screen | Formatted 2-column command reference table by display_help() |
| Bot Reply | Indented, bordered text block with 0.3 second delay by format_bot_reply() |
| Exit Condition | User types exit, quit, or bye — loop breaks and program closes |

### 4.2 Web Interface (Browser)

| Interface Element | Description |
|---|---|
| Chat Window | 420x620px white card, centered on screen with rounded corners |
| Header Bar | Dark green (#075e54) top bar with bot name and Online status |
| Quick Buttons | Pill-shaped shortcut buttons for Order Status, Payment, Offers, etc. |
| User Message Bubble | Green bubble (#dcf8c6) appearing on the right side of screen |
| Bot Message Bubble | White bubble appearing on the left side of screen |
| Typing Indicator | Three animated bouncing dots shown for 0.8 seconds before each reply |
| Input Bar | Text input at the bottom with a circular green send button |
| Enter Key Support | Pressing Enter triggers send — same as clicking the send button |

## 5. State and Session Management

The current chatbot is completely stateless. It does not remember previous messages or maintain any session context between turns. Every user message is processed independently as a fresh query.

| Aspect | Current Behaviour |
|---|---|
| Session Memory | None — each message is processed independently |
| User Identity | Not tracked — no login, username, or account system |
| Conversation History | Not stored — all chat data is lost when the session ends |
| Context Awareness | None — the bot cannot refer back to previous messages |
| Multi-turn Support | Not supported — each query is standalone |

Example of stateless limitation: If a user asks 'show my order status' and then follows up with 'what is the tracking number?' — the bot cannot answer because it has no memory of the previous exchange.

Future improvement: Maintaining a conversation_history = [] list would allow the bot to store and reference previous exchanges within a session.

# 6. Caching

No caching mechanism is implemented in the current version. Since the RESPONSES list is small (7 entries, approximately 2KB total) and is already loaded into RAM the moment the program starts, caching provides no additional benefit.

Every keyword search completes in under 1 millisecond — making caching unnecessary at this scale.

Future improvement: If the system were extended to fetch responses from an external database or REST API, caching frequently requested responses (such as shipping info or payment options) would significantly reduce response time and server load.

# 7. Non-Functional Requirements

## 7.1 Security Aspects

| Security Concern | How It Is Handled in This Project |
|---|---|
| Input Sanitization | User input is converted to lowercase and whitespace is stripped before any processing |
| No Personal Data Collection | The chatbot never asks for or stores names, passwords, card numbers, or any user data |
| No Network Exposure | The terminal version runs fully offline — zero network attack surface |
| XSS Prevention (Web Version) | JavaScript uses textContent instead of innerHTML — prevents script injection attacks |
| No File System Access | The program only reads from in-memory data — it cannot access or modify system files |
| No Third-Party Libraries | No external packages are used — eliminates supply chain security risks |

## 7.2 Performance Aspects

| Performance Metric | Value / Description |
| --- | --- |
| Response Time | Under 1ms — maximum 35 keyword checks (7 entries x 5 keywords) |
| Memory Usage | Less than 5MB — entire program and data fits comfortably in RAM |
| Startup Time | Under 1 second — no database connection, file loading, or network call required |
| Concurrent Users (CLI) | 1 user — single-threaded terminal application |
| Concurrent Users (Web) | Unlimited — index.html is a static file served directly by the browser |
| Scalability Limit | Up to 100 response categories before performance impact needs review |

# 8. References


- Python 3 Official Documentation — https://docs.python.org/3/
- MDN Web Docs (HTML, CSS, JavaScript) — https://developer.mozilla.org/
- Visual Studio Code Documentation — https://code.visualstudio.com/docs
- IBM HLD Format Template — provided by institution
- Project Source Code — ecommerce_chatbot/ folder (submitted separately)